Printed Pages – 14

576412 (76)
676413 (76)

MBA (4th Semester)

Examination, April-May, 2014

Branch : Management

ECONOMETRICS AND DECISION SCIENCE (NEW)

Time Allowed : Three Hours

Maximum Marks : 80

Minimum Pass Marks : 32

Note : Attempt any two from (a), (b) and (c). All the five questions are compulsory carrying equal marks.

Q. 1. (a) "Econometrics deals with the economic analysis with empirical approach". Explain the statement and mention the methodology of econometrics. 8

(b) Calculate the Gross Output of sector A, B and C for given data : **8**

Producing Sector	Receiving Sector			Final Demand	Total Output
	A	B	C		
A	30	80	40	50	200
B	10	35	25	130	200
C	25	15	30	30	100
Primary Input	135	70	5		

If the final demand of sector C changes to 30 to 40, what will be the output value of all sectors.

(c) For three sectors A, B and C based on the following data : **8**

Technology Matrix

	A	B	C
A	0.1	0.15	0.1
B	0.5	0.05	0.15
C	0.15	0.1	0.05

Output

A	100
B	200
C	300

(i) Prepare input-output transition table.

(ii) Find final demand of each sector.

(iii) Calculate primary inputs of each sector.

Q. 2. (a) Two competitors are competing for the similar product. The pay-off matrix in terms of their advertising plan is shown below : **8**

		Competitor B		
		Large advt.	med advt.	Small advt.
Competitor A	Large advt.	70	80	50
	Med advt.	90	60	95
	Small advt.	105	90	65

Find optimal strategies and value of the Game.

(b) Find the optimal strategies and value of the game using mixed strategy property. **8**

Player B

		1	2	3
Player A	1	3	–1	4
	2	6	7	–2

(c) Find the optimal strategies and value of the game using graphical method. **8**

Player B

		B_1	B_2
	A_1	3	4
	A_2	–3	12
Player A	A_3	6	–2
	A_4	–4	–9
	A_5	5	–3

Q. 3. (a) "The Markov analysis may be understood as the way of analysing the past and present movement of some variable in an effort to forecast the future movement of that variable". Explain this statement by taking examples from marketing and personnel. **8**

(b) The arrival behaviour of bank employees reveals that if he is late on a day, he is 90% sure to be in-time the next day. Similarly if he is on-time on a day there is a 30% chance that he will be late the next day. How often in the long run is he expected to be late for the office. **8**

(c) To investigate brand switching between different brands of detergent powder, a research company surveyed shoppers in an area to discover their behaviour. The survey on a number of shoppers in relation to three leading brands superwash, clean and shine has revealed the following information : 8

Current Buy	Previous Buy	Number
Super wash	Super wash	200
Clean	Clean	150
Shine	Shine	100
Clean	Super wash	50
Shine	Super wash	25
Super wash	Clean	80
Shine	Clean	45
Super wash	Shine	130
Clean	Shine	20

Calculate the market share for each brand in the long run.

Q. 4. (a) Explain and illustrate the following principles of decision-making : 8

(i) Laplace

(ii) Maximin

(iii) Maximax

(iv) Hurwicz

(v) Savage

(b) Informatics corporation summarises international information reports (on a weekly basis) prints sophisticated data and forecasts, which are purchased weekly by

mutual funds, banks and insurance companies. This information is very expensive and the demand for the reports is limited to a maximum of 30 units. The possible demands are 0, 10, 20 and 30 reports per week. The profit per report sold is Rs. 30 and the loss per report unsold is Rs. 20. No production of extra reports during a week is possible. Futher there is a penalty cost of Rs. 250 for not meeting the demand. Unsold report can not be carried on to the next week. Using the pay-off table, find out the number of reports to be produced if : **8**

(i) Maximin or pessimistic strategy is adopted.

(ii) Maximax or optimistic strategy is used

(c) An oil company has recently acquired rights in a certain area to conduct surveys and test drillings to lead to lifting oil if it is found in commercially exploitable quantities.

The area is considered to have good potential for finding oil in commercial quantities. At the outset the company has the choice to conduct further geological tests or to carry out a drilling programme immediately on the known conditions the company estimates that there is a 70 : 30 chance of further tests showing a 'success'.

Whether the tests shows the possibility of ultimate success or not or even if no tests are undertaken at all, the company could still pursue its drilling programme or alternatively consider selling its rights to drill in the area. Thereafter however if it carries out the drilling programme the likelihood of final success of failures is considered dependent on the foregoing stages thus

→ if 'successfull' tests have been carried out, the expectation of success in drilling is given as 80 : 20

→ if the tests indicate 'failure' then the expectation of success in drilling is given as 20 : 80

→ if no tests have been carried out at all, the expectation of success in drilling is given as 55 : 45

Costs and revenues have been estimated for all possible outcomes and the net present value of each is as follows : 8

	Outcome	Net present value (Rs. Million)
Success :	With prior tests	100
	Without prior tests	120
Failure :	With prior tests	–50
	Without prior tests	–40
Sale of exploitation rights :		
	Prior test show 'success'	65
	Prior test show 'failure'	15
	Without prior tests	45

Find :

(i) Draw up decision (probability) tree diagram to represent the above information; and

(ii) Evaluate the tree in order to advise the management of the company on its course of action.

Q. 5. (a) What are the steps involve in simulation ? Explain with suitable examples. **8**

(b) Discuss Monte Carlo simulation. Illustrate how would you use it in situations of : **8**

(i) Queuing and

(ii) Inventory control

(c) Dr. Strong is a dentist who schedules all her patients for 30-minutes appointments. Some of the patients take more or less than 30 minutes depending on the type of dental work to be done. The following summary shows the various categories of work their probabilities and the time actually needed to complete the work : **8**

Category	Time Required	Probability of Category
Filling	45 minutes	0.40
Crown	60 minutes	0.15
Cleaning	15 minutes	0.15
Extraction	45 minutes	0.10
Check-up	15 minutes	0.20

(14)

Simulate dentist's clinic for four hours and determine average waiting time for the patients as well as the idleness of the doctor. Assume that all the patients show up at the clinic at exactly their scheduled arrival time starting at 8 a.m. Use the following random numbers for handling the above problem :

40, 82, 11, 34, 25, 66, 17, 79